

Finding Object Depth Using Stereoscopic Photography^{*}

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Abstract

Stereoscopic scenes of the mankind is naturally caused by synthesizing two images produced by the parallax of the two eyes of human. Such being the case, mankind can distinguish the relative position of the objects. In the study of related stereovision, some persons aim at the framework of taking simulated images with two eyes using one or two cameras from front and back or simultaneously at the same time to obtain a pair of parallax mages; someone pay more attention to the theoretical analysis of the relative positions of the parallax images, and some others do the work of using parallax images as material to carry out the job of image classification, comparison and analysis. The purpose of this paper is to estimate the distance between the objects and the camera in an outdoor environment using commercial digital camera.

Key words: computer vision, machine vision, stereoscopic, object depth, binocular disparity, depth perception.

1. Introduction

Human can read stereo information because they can see two overlapping views from two eyes [8]. Unlike most other animals, human have two eyes in the front of their face, it means that they have lost the field of vision but increased the ability of judgment in object depth and distance. The average distance between the two eyes is about 6.5 cm that makes human capable of distinguishing the object from background when watching the object nearby. Binocular disparity is a phenomenon caused by two different images viewed by two eyes. Binocular disparity is closely related to the distance between camera and object.

Some stereo analysis assumes an ideal pin-hole camera model which offers an infinite depth of field. However, any practical camera system will produce depth-related blurring. Stereo images are usually obtained by using two cameras separated by a small disparity distance. However, the difference of system

parameters between these two cameras and the difficulty in setting up the camera pair, make it an arduous task to obtain well-controlled stereo images. Single camera stereo has several advantages over traditional two-camera stereo, because under single sensory condition, system parameters such as spectral response, gain, and offset are identical for the stereo pair. In addition, data acquisition is greatly simplified because no hardware or software is needed for synchronization and only a single set of internal parameters needs to be calibrated [7].

The horizontal separation of the two eyes gives rise to small positional differences, called binocular disparities, between corresponding features in the two retinal images. Each eye sees a slightly different view of the world due to the horizontal separation of the two eyes.

Lee et al. [4] presented that stereo images are usually obtained either by displacing a single camera in the scene or using two cameras mounted on a platform and separated by a small distance. The stereo disparity between the two images from two distinct viewpoints is a powerful cue to 3-D shapes and poses estimation.

Stereo analyses defined by Barnard and Fischer [11] include six necessary steps as follows: image acquisition, camera modeling, feature acquisition, image matching, depth reconstruction, interpolation. Lavoie et al. [10] proposed that these steps can be further grouped into three more general stages, namely: (1) Preprocessing which involves finding of the global position, orientation and the calibration of the cameras. (2) Correspondence finding, which establishes a correspondence between points from the two images, is traditionally done by using two different methods: area-based and feature-based matching. (3) Depth calculation. Lee et al. [4] proposed that the stereo process can be summarized by the following steps: (1) detection of features in each image; (2) matching of features between the images under certain geometric and other constraints; and (3) calculation of depth using the disparity values and the geometric parameters of the imaging

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configuration. Epipolar geometry [3, 9] has been a convenient tool for description and analysis of multi-view vision system. It is used to represent the geometric relationship between two viewpoints of the same scene based on a few corresponding points in a pair of images.

In this paper, we develop the usual procedures to show how to take stereoscopic photograph by using the digital camera without additional parts. The purpose of this paper is to estimate the distance between the objects and the camera in an outdoor environment using commercial digital camera. We proposed an approach to rapidly get the parameters of digital camera for different brands, and we designed the calibration on the outsides of the digital camera using least square regression method to find the focus. The optimum stereo base line is essential for perfect stereoscopic photograph. Further, we can estimate the distance of the object by stereoscopic photograph.

The rest of this paper has been organized as follows. In Section 2, we will discuss about stereoscopic photograph we had previously studied. In Section 3, we will present the main ideas of this procedure, and the depth perception methodology that we used. The optimum stereo base line calculation methodology for each case is briefly exposed. In Section 4, some experimental analysis and results are presented. The conclusions and contributions of this paper are stated in Section 5.

2. Previous Studies

Stereo binocular disparity presents images and provides information about depth for human. In the computer vision research area, it's important to recover the 3D information of a scene from 2D images. In respect of architectural stereo system, some designs use one or two cameras and some combine with other equipment, such as biprism. Each has its merits. In this research, we use one camera to take the photograph twice in slide plate with bar. It means that we can make work down in low cost. The previous work [4, 7] about architecture discusses as follows and shows in Table 1 and Figure 1. We classified them into three aspects: rotation of the parts, two/one shots of a scene and a static/dynamic scene in use.

A single camera not only provides radiometric advantages over traditional two camera stereo, but also reduces the complexity and cost of acquiring stereoscopic video. Wu et al. [7] proposed to use planar mirror catadioptric stereo systems to capture stereo images with only a single camera so that the necessity of camera calibration could be avoided. On the basis of affine epipolar geometry, a projective rectification scheme can be replaced by a similarity transformation. It also demonstrated that the rectification algorithm is very effective for rectifying

planar catadioptric stereo imagery.

Lee et al. [4] by using a single-lens camera system to obtain stereo images in one picture that can get more profit, such as unwanted geometric and intensified differences between the stereo images are reduced. It is also easy to set up the biprism so that the corresponding features lie on the same scan line automatically.

Stereo analysis can be grouped into three general stages, preprocessing, correspondence finding and depth calculation. The previous work about stereo analysis discusses as follows. In respect of preprocessing, Marr and Poggio [5] proposed a co-operative computation stereo algorithm based on a multi-resolution framework. Recently scientists start to look at depth perception when viewing renderings of 3D environments. Lol et al. [2] showed great promise for achieving true realism but if should be achieved by considering stereovision which is the dominant human viewing experience. It has been proposed that viewing in stereo will affect our judgment of the visual realism of the computer-generated images. Moreover, from a computational point of view, by perceiving real depth from binocular disparity we might be able to reduce the computation of monocular depth cues such as shadows, while maintaining the same perceptual response [7].

From correspondence finding point of view, Julesz [3] showed that random dot stereo grams provide a cue for disparity even when each image does not provide any high-level cue for depth. Pentland [1] reported a finding that the gradient of focus inherent in biological systems is also a useful source of depth information. Stereo correspondence research has traditionally been, and continues to be, one of the most active investigated topics in computer vision. They present the exhaustive surveys on the stereo correspondence algorithms. Also compare the existing stereo methods and present experiments evaluating the performance of many different variants.

Let us consider the depth calculation. Barnard and Thompson [12] proposed a feature-based iterative algorithm to solve the disparity related problems. Lavoie et al. [10] noted a new method for reconstructing 3-D images from the "left" and "right" two-dimensional (2-D) images of an object using a pseudorandom encoded structural light and introduced it for use. The depth of each point on the object surface is calculated by using the disparity map for calculating the "right" and "left" images. When two images of an object with a PRBA (pseudorandom binary array) are projected onto it and taken, a disparity map can be easily obtained for the two images. In our research, we consider the subject under the technology of stereoscopic photography to distinguish the object depth applied

to outdoor scene. We are interest in estimate of camera parameters and depth calculation.

3. The Geometry of Multiple Views

In Figure 2(a) we can see that the signs appearing in the rear of the scene is tended to left just like human's left eye which has observed it or otherwise vice versa as shown in Figure 2(b). The phenomenon of binocular parallax is the principal theory of stereoscopic photography. We put the camera in two different places and took the picture in turn. In other words, we can take a pair of pictures to simulate binocular parallax effect. Binocular parallax effect depends on the distance between two shots which we call it as stereo base line. We can get that the longer the stereo base line the better the stereo effect for the view of a far way mountain.

Some new and advantageous photographic technologies are being introduced such as image base virtual reality, panama, and stereo photo. Many of them are being implemented in the commercial digital camera, with the only exception of stereo photo efficacy, which is being implemented only in some types of Pentax cameras. We proposed a simplified method of taking stereo photo by introducing the use of only one commercial digital camera. We propose two stages of works in our experiment. First stage, we can take serial photos with regular displacement along the slide bar and make preprocesses if necessary. The steps of taking stereo photo are showed as follows.

1. Estimate the distance of the main objects which you want to take the photo.
2. Determine the horizontal distance of the camera between two shots.
3. Carry out post processing to combine left and right image and take a look with accessories.

Second stage, according to the different purposes, we can divide the works into two parts. First part, for entertainment we can select photo-pairs and use stereoscopic glasses to present with 3D scene. Second part, for application we propose the model of lineal regression after training and validation. We can use the results in real world for depth estimation.

The X-Z plane coordination of our proposed stereoscopic photograph camera system is shown in Figure 3. Many important parameters are also shown in this Figure.

$$x_l = \frac{fx}{z} \quad (3.1)$$

$$x_r = \frac{f(x-b)}{z} \quad (3.2)$$

From Eq. 3.1 and Eq. 3.2, we obtain Eq. 3.3, as follows.

$$\begin{aligned} x_l - x_r &= \frac{fx - f(x-b)}{z} \\ z &= \frac{fx - f(x-b)}{x_l - x_r} \\ z &= \frac{fb}{d} \end{aligned} \quad (3.3)$$

where $d = x_l - x_r$, is the disparity of two corresponding point.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{x_l z}{f} \\ \frac{y_l z}{f} \\ \frac{fb}{x_l - x_r} \end{bmatrix} = \begin{bmatrix} \frac{x_l z}{f} \\ \frac{y_l z}{f} \\ \frac{fb}{d} \end{bmatrix} \quad (3.4)$$

Depth (z) goes to infinite as disparity (d) approaches to zero. As shown in Eq. 3.5 we can see that depth(z) is increased while disparity(d) is down if base line(b) keeps a constant. We can use this property to estimate the focus of the camera by Least Square Method.

$$z = \lim_{d \rightarrow 0} \frac{fb}{d} = \infty \quad (3.5)$$

4. Experimental Method and Results

In this paper, general commercial digital camera such as Casio Z4, or special camera supported by half-frame shot function like Pentax S5i which is also called stereoscopic module is used. The camera attached to slide plate with bar so that the displacement is accurate. We also need stereoscopic glasses viewer to look at the photograph which we get and make sure that the scene of the stereo photo is properly attractive.

First, we marked the objects which were dispersed in the scene as shown in Figure 4. We used photo the odolite to measure the absolute coordinates of the markers as shown in Table 2. In Figure 4, we selected and labeled eight points in the scene. In Table 2, we showed the absolute coordinates of each point.

Second, we increased b value in regularity between one and one-hundred centimeter to take picture for analysis. We took one hundred and one pictures in each run. The numbers of the data records in b value are shown in Table 3.

For comparison, we used two kinds of digital cameras to offer the different resolutions in our experiment. We also selected a number of different outdoor and indoor scenes for evaluation.

In our experimental design for each run we can get ninety-six pictures in different baseline at least,

so we can calculate b value from one to ninety-five centimeter. Count of the data records in different b value is shown in Table 3. On the other hand, we care very much about the confidence of the analysis, so we use b value within one to eighty-one for data analysis. That means the data records less than twenty will be omitted in analysis.

In this section, we tried to find out proper regression equation and how many points we need to train. We also tried to calculate the predicted error rate in validation sets.

$$\text{Predicted error rate} = \frac{|(\text{estimated } z - \text{true } z)|}{(\text{true } z)} \quad (4-1)$$

We separated six points into two portions; one of them was called training data sets and the other called validation data sets. In the training data sets we calculated the parameter of regression as shown in Table 4. In Table 4, we can see that the average numbers of parameters are quite similar and lay between 25.254 (minimum) and 26.291 (maximum). In Table 5, we used analysis of variation (ANOVA) to test significance. We found that F value was less than critical value in 0.05 level of signification, so we accepted the null hypo-paper. It means that the numbers in the training set are not significantly different.

In Table 6, we can find that parameter different ratios all lie in between -2% and 2%. We put point 2, 3, 4, 5 into training data set and the result of least square regression in SPSS are shown in Table 7.

In Table 9, we can find the regression equation as follows, and we use it to estimate the object distance. For Pentax, the regression equation is $z_d = 25.637 \times b$, and focus is 25.637. For Casio, the regression equation is $z_d = 24.028 \times b$, and focus is 24.028. And we put the other point 2' and 5' into validation data sets to predict the depth of the points and calculate the predicted error rate as shown in Table 8. In Table 8, we can find the predicted error rate between training and validation is quite near.

We used different base line within one to eight to take pictures in indoor. In outdoor place, we put three markers in 5m, 10m and 15m. We used two kinds of cameras and different base line within five to thirty-five to take pictures. Besides we used the parameter we found in Section 4.4 to estimate the distance of markers as shown in Figure 5 and 6. The results are shown in Table 9 and 10.

5. Conclusions

From what we have said above, we can come to the conclusion and summarize the statements as follows:

1. Concerned experts suggest that the base line is equal to one fortieth or one fiftieth of the distance between the primary object and camera [12, 14]. From our experiment we find base line is not a sensitive parameter in stereoscopic photograph, but

the farther the object the more the base line is the rule.

2. In Table 5, we used analysis of variation (ANOVA) to test significance. We find that F value is less than critical value in 0.05 level of signification, so we accept the null hypo-paper. It has been confirmed that the proper numbers in the scene are not significantly different.

3. The Eq. 3.3 is in very good agreement with the experiments as shown in Figures 4. We can calibrate the camera easily as shown in Section 4. In Section 4 we can find that the different digital camera has different focus, in our experiment for Casio-4z it is 24.028 and for Pentax-S5i it is 25.637. The equations of regression are $z_d = 25.637 \times b$ for Pentax-S5i and $z_d = 24.028 \times b$ for Casio-4z. We can use the equations to estimate the distance from camera to object in the scene.

4. In Table 8, we found that the predicted error rate between training and validation are quite near. The error rate of validation points is between 1.2% and 2.7%.

5. In Table 9, we find that the results of estimate in Figure 5 are reasonable. But further points are unexpected, because b value within one to eight for far objects may cause bigger error. In Table 10, we used b value within five to thirty-five and find that the predicted error rate lie in between 0.45% and 5.96%.

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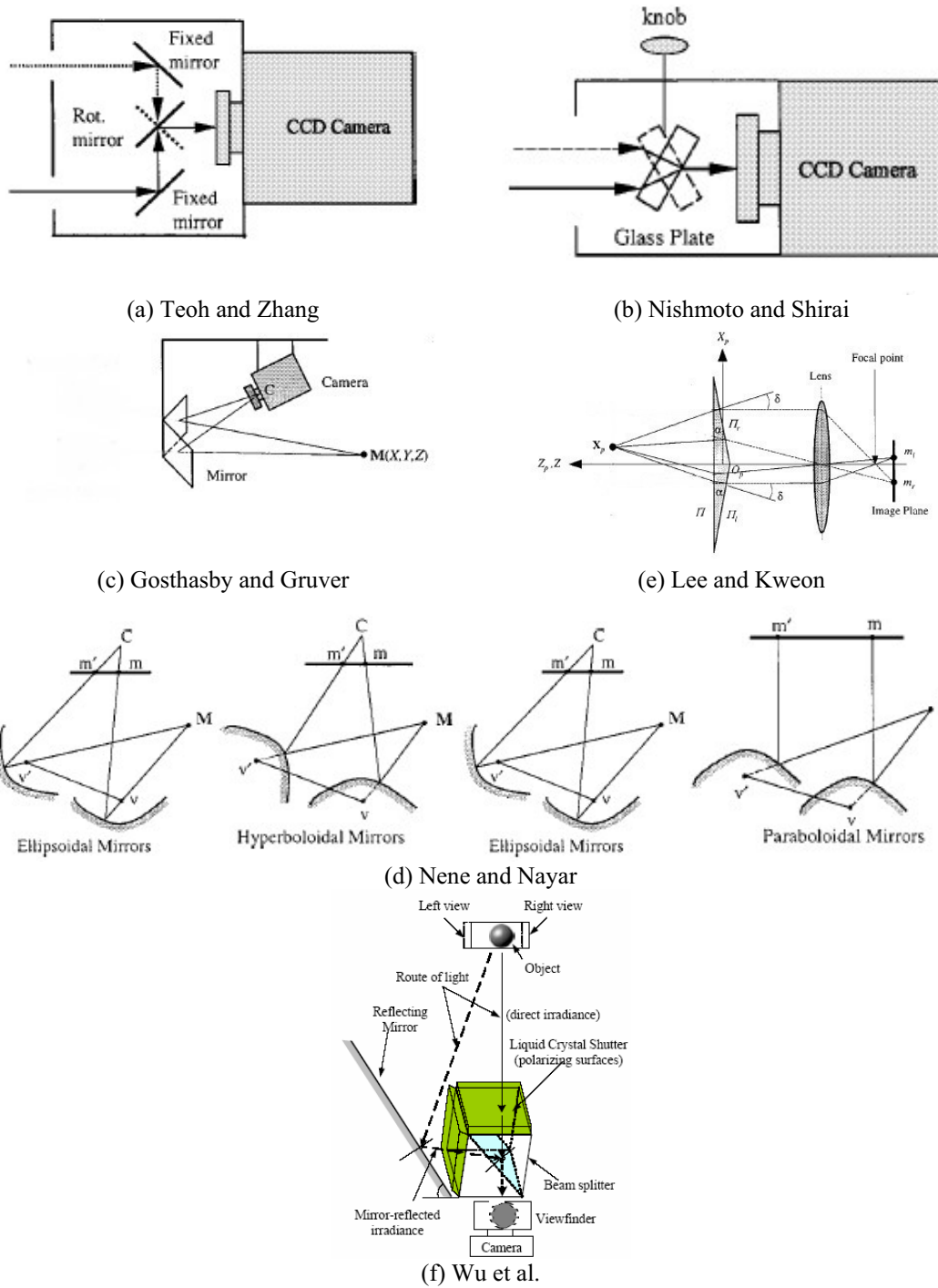


Figure 1. Architecture to obtain stereo images with a single camera



Figure 2. A demonstration of binocular parallax. (a) Left image frame. (b) Right image frame.

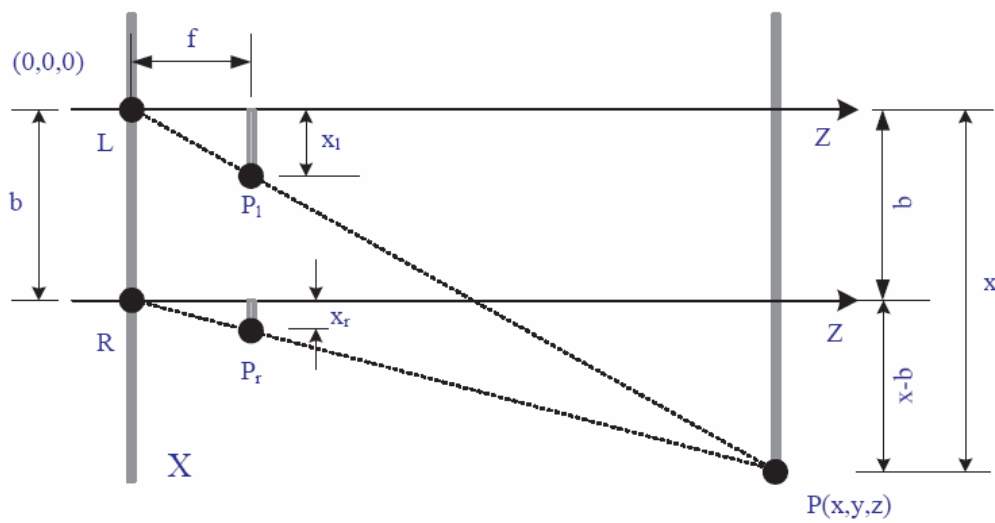


Figure 3. Coordinate system in X-Z plane.



Figure 4. The marker was dispersed in the scene. (a) The marker for phototheodolite to measure the distance. (b) The scene we took when the camera attached to slide plate with bar in 5 cm. (c) The scene we took when the camera attached to slide plate with bar in 105 cm.

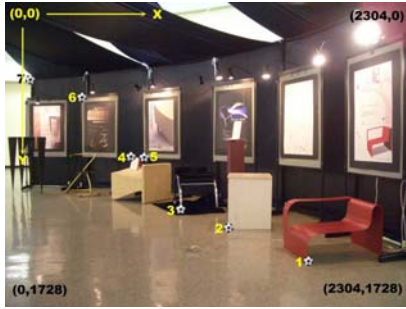


Figure 5. Distance estimated in indoor scene.



Figure 6. Distance estimated in outdoor scene.

Table 1. Using a single camera to obtain stereo image with different behaviors

Item	Rotation of the parts	Two/one shots of a scene	Static/dynamic scene in use	Figure 1
Teoh and Zhang	rotation	two	static	(a)
Nishmoto and Shirai	rotation	two	static	(b)
Gosthasby and Gruver	No rotation	one	static/dynamic	(c)
Nene and Nayar	No rotation	one	static/dynamic	(d)
Lee and Kweon	No rotation	one	static/dynamic	(e)
Wu et al.	rotation	two	static	(f)

Table 2. The absolute coordinates of the markers.

coordinate	x(m)	z(m)	y(m)
1	-2.131	3.724	1.484
2	-2.006	6.398	1.115
2'	-1.416	6.398	1.115
3	0	10.631	1.118
5'	1.436	6.977	1.161
4	1.519	11.054	1.502
5	2.056	6.977	1.118
6	1.998	4.224	1.286

Table 3. The numbers of the data records in different b value.

b	records	b	records	b	records	b	records	b	records
1	100	21	80	41	60	61	40	81	20
2	99	22	79	42	59	62	39	82	19
3	98	23	78	43	58	63	38	83	18
4	97	24	77	44	57	64	37	84	17
5	96	25	76	45	56	65	36	85	16
6	95	26	75	46	55	66	35	86	15
7	94	27	74	47	54	67	34	87	14
8	93	28	73	48	53	68	33	88	13
9	92	29	72	49	52	69	32	89	12
10	91	30	71	50	51	70	31	90	11
11	90	31	70	51	50	71	30	91	10
12	89	32	69	52	49	72	29	92	9
13	88	33	68	53	48	73	28	93	8
14	87	34	67	54	47	74	27	94	7
15	86	35	66	55	46	75	26	95	6
16	85	36	65	56	45	76	25	96	5
17	84	37	64	57	44	77	24	97	4
18	83	38	63	58	43	78	23	98	3
19	82	39	62	59	42	79	22	99	2
20	81	40	61	60	41	80	21	100	1

Table 4. Regression analysis with different number of training data sets.

number of select point	1	2	3	4	5	6
Parameter of regression	25.265	25.586	25.757	25.891	25.896	25.791
	25.907	25.683	25.700	25.800	25.768	
	26.100	25.778	25.821	25.632	25.729	
	26.291	25.596	25.700	25.896	25.691	
	25.927	25.260	25.475	25.728	25.763	
	25.254	26.004	25.885	25.684	25.898	
		26.099	25.764	26.056		
		25.917	25.540	25.888		
		25.581	25.828	25.637		
		26.195	25.603	25.845		
		26.014	25.482	26.056		
		25.677	26.099	25.888		
		26.109	25.978	25.797		
		25.772	25.754	25.845		
		25.591	26.042	25.893		
			25.817			
			25.696			
			26.106			
			25.882			
			25.824			
average	25.791	25.791	25.788	25.8357	25.791	25.791
median	25.917	25.772	25.791	25.8447	25.765	25.791
Standard deviation	0.4341	0.2594	0.1828	0.12859	0.0868	
variance	0.1884	0.0673	0.0334	0.01654	0.0075	
minimum	25.254	25.26	25.475	25.6316	25.691	25.791
maximum	26.291	26.195	26.106	26.0563	25.898	25.791
samples	6	15	20	15	6	1

Table 5. ANOVA with different number of training data sets.

variables	SS	df	MS	F	P-value	Critical value
Between-class	0.024518	5	0.004904	0.100234	0.991678	2.376684
Within-class	2.788551	57	0.048922			
sum	2.813069	62				

Table 6. Different ratio with different number of training data sets.

	point of no. we selected	Parameter of regression	different ratio with all selected		point of no. we selected	Parameter of regression	different ratio with all selected		point of no. we selected	Parameter of regression	different ratio with all selected
1	5'	26.291	-1.9%	3	35'4	26.106	-1.2%	4	2'35'4	26.056	-1.0%
	3	26.100	-1.2%		2'35'	26.099	-1.2%		235'4	26.056	-1.0%
	4	25.927	-0.5%		2'5'4	26.042	-1.0%		22'5'4	25.896	-0.4%
	2'	25.907	-0.5%		2'34	25.978	-0.7%		35'45	25.893	-0.4%
	2	25.265	2.0%		235'	25.885	-0.4%		22'35'	25.891	-0.4%
	5	25.254	2.1%		35'5	25.882	-0.4%		235'5	25.888	-0.4%
2	35'	26.195	-1.6%	3	25'4	25.828	-0.1%	4	2'35'5	25.888	-0.4%
	5'4	26.109	-1.2%		5'45	25.824	-0.1%		25'45	25.845	-0.2%
	2'5'	26.099	-1.2%		22'5'	25.821	-0.1%		2'5'45	25.845	-0.2%
	34	26.014	-0.9%		2'5'5	25.817	-0.1%		22'34	25.800	0.0%
	2'3	26.004	-0.8%		234	25.764	0.1%		2'345	25.797	0.0%
	2'4	25.917	-0.5%		22'3	25.757	0.1%		22'5'5	25.728	0.2%
	25'	25.778	0.1%		2'35	25.754	0.1%		22'45	25.684	0.4%
	5'5	25.772	0.1%		22'4	25.700	0.4%		2345	25.637	0.6%
	23	25.683	0.4%		22'4	25.700	0.4%		22'35	25.632	0.6%
	35	25.677	0.4%		2'45	25.696	0.4%		22'35'4	25.898	-0.4%
	24	25.596	0.8%		25'5	25.603	0.7%		2'35'45	25.896	-0.4%
	45	25.591	0.8%		235	25.540	1.0%		235'45	25.768	0.1%
	22'	25.586	0.8%		245	25.482	1.2%		22'35'5	25.763	0.1%
	2'5	25.581	0.8%		22'5	25.475	1.2%		22'5'45	25.729	0.2%
	25	25.260	2.1%						22'345	25.691	0.4%

Table 7. The result of least square regression in SPSS.

camera	Unstandardized Coefficients		t	Sig.
	b	Std. Error		
Pentax	25.637	.009	2888.383	.000
Casio	24.028	.020	1179.23	.000

Table 8. Predicted error rate with 4 points training set.

No. of point	2	2'	3	5'	4	5
Average of predicted value	6.479	6.323	10.401	6.790	10.905	7.068
True value	6.398	6.398	10.631	6.977	11.054	6.977
Predicted error rate	1.3%	1.2%	2.2%	2.7%	1.3%	1.3%

Table 9. The result of estimate for some points in indoor scene.

b(cm)	No.1	No.2	No.3	No.4	No.5	No.6	No.7
1	3	0	3	7	3	4	0
2	9	6	9	10	6	3	6
3	12	16	12	13	3	6	9
4	22	19	16	16	9	9	9
5	25	19	19	22	16	12	12
6	38	28	25	22	19	22	22
7	38	28	31	25	22	22	22
8	44	34	31	35	22	25	22
Distance estimated	4.20	5.61	6.19	7.08	8.04	7.35	7.80

Table 10. The result of estimate for some points in outdoor scene.

Camera b(cm)	For casio			For pentax		
	No.1	No.2	No.3	No.1	No.2	No.3
5	23	11	8	25	12	8
10	47	24	17	50	24	16
15	71	37	26	76	37	25
20	96	50	34	103	51	34
25	119	62	43	127	62	41
30	140	71	50	150	73	48
35	160	80	55	174	84	54
Distance estimated	5.143	10.045	14.484	5.095	10.429	15.895
Predicted error rate	2.87%	0.45%	3.44%	1.89%	4.29%	5.96%